HLD

High-Level Design Document (HLD) - Sentinel AI

> Deliverable 2. End-to-end architecture, pipelines, schemas, integration, security, HA/DR. Companion to `01_Detailed_Technical_Design.md` (984 lines) - this HLD is the evaluation-ready concise view that directly answers each required bullet.

1. End-to-End System Architecture & Component Interactions

Hybrid Model: Middleware Federation (Model 3) + Central VMS (Model 4) + GIS Registry (Model 1). See `05_Architecture_Principles_Compliance.md` and `Combined_Models_Integration_Architecture.md`.

```
[ 50 Heterogeneous Cameras - 7 vendors, 19 districts, 4 boards ]
```

```
Additive Alembic chain `0001->0006->0008->0009`; federation owned by `FederationBase.metadata.create_all` at runtime (not alembic) - `apps/api/src/federation/db.py`.
```

2. Network Topology

- **Edge:** District PoPs aggregate RTSP (10.10.0.0/16 subnets per cluster; `10.10.1.x` AHM, `10.10.3.x` SRT, etc.). MediaMTX per PoP in 80k design; today single `sentinel-mediamtx`.

3. Ingestion Pipeline & Streaming Protocols

| Stage | Protocol | Config |

4. Storage Tiering (Hot / Warm / Cold)

| Tier | Store | Retention | Content |

PostGIS not required for reads (portable floats + pure-Python GIS math in `registry/gap.py`); `Geometry` column available when enabled.

5. Database Schemas

5.1 Camera Metadata

- `cameras(id, name, rtsp_url, location, latitude, longitude, status, is_active, last_seen_at)` - `src/models/camera.py`

5.2 Event Logs

- `PlateDetection(camera_id, camera_name, plate, normalized_plate, state_code, rto_code, ocr_confidence, detection_confidence, vehicle_type, color, make/model, x/y/w/h, backend sim, frame_seq, ts)` - 8 rows (2 plates).

5.3 Watchlist Records

- `Watchlist(identifier_number UQ, target_type{vehicle/person}, category{stolen/wanted/missing/blacklisted/suspect}, source_db{VAHAN/eGujCop/SARTHI/INTERNAL}, notes, active, added_by FK)` - 10 rows, 9 active; lookup `GET /watchlists/lookup/{plate}`.

5.4 Federation IAM (fed_* - 17 tables, 14 depts)

`fed_departments (GJPOL state_hq + descendants), fed_roles (21), fed_permissions (320), fed_role_permissions, fed_officers (ADMIN-0001 super_admin), fed_officer_sessions, fed_audit, fed_secrets, fed_retention, fed_webhooks, fed_camera_groups, fed_department_cameras, fed_cameras, fed_streams, fed_health, fed_emergency_access, fed_abac_policies, fed_officer_sessions` - `src/federation/models.py`; persisted via `FED_DATABASE_URL` to same Postgres (not in-memory).

6. Integration Strategy for Departmental Systems

| Interface | Standard | Endpoint |

7. Cybersecurity, Encryption, RBAC, Compliance

- **RBAC + ABAC/Jurisdiction:** 21 roles, 320 permissions, `RBAC`, `ABAC`, `Isolation`, `Jurisdiction` (`src/federation/security/*`); `_scope_filter` enforces department isolation; department_admin scoped to own `TRAF` etc.

8. Disaster Recovery, Redundancy, High Availability

- **HA:** Stateless API (HPA), Redis cluster, Postgres streaming replication (primary + replicas), MediaMTX per district (80k). Cluster plane `src/cluster/router.py` (`register, heartbeat, lease, failover, ownership`).

9. Deployment (Evaluation)

`docker compose up --build -d` (api, web, postgres postgis:16-3.4-alpine, redis, mediamtx) -

`FED_DATABASE_URL=postgresql+asyncpg://sentinel:***@postgres:5432/sentinel` (persisted), `STREAM_ALLOW_TEST_SOURCES=true` for `lavfi` injection, `SEED_ON_STARTUP=true` -> 51 cameras/51 registry/10 watchlists/8 ANPR hops. Health `GET /api/v1/health` (db/redis latency ms), Web `http://localhost:3000/login`.

10. Traceability to Code

| HLD item | File |